

Public Summary

When Memory Gets Stuck or Slips Away: A Brain Circuit Story

A plain-language summary of: "NMDA Controls Working Memory Attractor Stability: Opposite Regimes Produce Schizophrenia-like Instability and OCD-like Overstability" — Hamza TAHIRI, Fatemeh AMIDIZADE, Razieh ZARE, Ibtissam CHAKIR, Arvind KUMAR, April 2026

The everyday problem

Imagine you look up a phone number, walk across the room to make the call, and — it is gone. Or imagine the opposite: a thought lodges itself in your mind and you simply cannot let it go, no matter how hard you try.

These are not just frustrating moments. For millions of people living with schizophrenia or OCD, they are daily reality. What if both problems — memory that vanishes too easily, and memory that refuses to leave — were actually two sides of the same coin?

This is exactly we are trying to show.

What is working memory?

Working memory is akin to a mental scratchpad. This is what lets you hold a thought in mind when there is no longer anything in front of you to look at, during the pause between hearing instructions and carrying them out, or between seeing a face and remembering the name that goes with it.

How the brain creates and maintain working memory is an active field of reserach and some important clues have emerged. For instance, small groups of neurons in the prefrontal cortex keep firing even after the original signal (input) is gone, like an echo that sustains itself [10,11]. Neuroscientists refer to this as the **persistent activity**, and it depends on a careful balance of signals between brain cells [2,3].

The key molecule: NMDA receptors

At the heart of this balance is a type of molecular gateway called the NMDA receptor. These receptors at the post-synaptic spines not only mediate the flow of excitatory signals but also allow the influx of calcium, which ultimately controls the strengthening or weakening of synaptic connections [12]. In addition, the excitatory currents carried by NMDA receptors have an unusually long time constant, far slower than the fast AMPA receptor currents that dominate moment-to-moment signalling, which allows them to integrate past neural activity over hundreds of milliseconds. It is precisely this property that makes NMDA receptors a critical candidate for sustaining working memory: they act as a slow, self-amplifying echo of recent activity, keeping the memory trace alive long after the original input has gone [1,2].

The level of NMDA conductance therefore sets how deep the memory "well" is? how strongly the circuit resists being knocked out of its active state by noise or competing inputs. Too little NMDA and the well becomes too shallow; too much and the circuit becomes trapped in it. This gives NMDA the character of a bidirectional gain control on memory stability, with three qualitatively distinct regimes:

- **Too quiet (low NMDA):** The self-sustaining echo fades. The attractor basin is too shallow to resist neural noise, and the memory collapses back to baseline.
- **Just right:** The memory holds steadily across the delay, flexible enough to be updated by new information, yet stable enough to last until it is needed.
- **Too loud (high NMDA):** The echo locks in and won't stop. The attractor becomes so deep that normal cognitive signals cannot clear it, and thoughts become "sticky" and perseverative.

What this research did

We built a **computer model of a brain circuit** — our model consists of 2,000 neurons wired together the way they are in the prefrontal cortex, with biologically plausible electrical properties and three types of synaptic connections (AMPA, NMDA, and GABA), following the landmark spiking network framework of Wang (2001) and Deco & Rolls (2007) [2, 6].

We then simulated *in silico* experiments:

1. **Showed the simulated network a cue** (a burst of input to one group of neurons)
2. **Removed the cue** and watched whether the memory held during the delay
3. **Introduced a distractor** (a competing signal) to test robustness
4. **Measured a new metric** called λ_{delay} : a number that captures how quickly the memory trace decays. Low λ = strong memory. High λ = memory collapses.

We tested this across three versions of the circuit:

Version	NMDA level	Mimics
Control	Normal (100%)	Healthy brain
SCZ-like	Reduced (−5%)	Schizophrenia
OCD-like	Elevated (+10%)	OCD

What we found

We made a number of observations that reveal the important role of NMDA in working memory and thereby in the symptoms of schizophrenia and OCD like brain disorders:

In the healthy circuit, memory held steadily across the delay. The neurons kept firing at around 20 Hz: a sustained, selective 'hum' that represented the remembered item. $2 < \lambda < 5$ (strong persistence).

In the OCD-like circuit, memory became almost impossible to extinguish. Even at weaker synaptic connections, the circuit got locked into high activity and refused to return to baseline. $\lambda < 2$ (over-stabilised). The mental scratchpad got stuck.

In the schizophrenia-like circuit, the memory collapsed almost immediately after the cue disappeared. The echo faded before the delay even ended. $\lambda > 5$ (near-total failure). The mental scratchpad couldn't hold anything.

Crucially, we showed this across a whole range of synaptic strengths — not just at one setting. The schizophrenia-like circuit almost *never* reached the persistence zone. The OCD-like circuit entered it *much earlier and stayed there* far more broadly.

Why this matters

To the best of our knowleged, this is the first time both disorders have been placed on the **same mechanistic axis**, not as separate disease processes, but as opposite ends of the same dial.

For schizophrenia: The model supports the hypothesis that NMDA hypofunction (well-documented post-mortem and in imaging studies [4] and famously reproduced by sub-anaesthetic doses of ketamine in healthy volunteers [13]) makes prefrontal memory states fragile and easy to disrupt,

directly explaining the well-known working memory deficits and hypofrontality [5] and the reduced prefrontal signal-to-noise ratio seen in patients [9].

For OCD: The model offers a biophysical account of what "getting stuck in a thought" actually means at the neural level, an attractor basin so deep that normal updating signals can't push the circuit out of it.

For treatment: The work suggests that therapeutic interventions targeting NMDA (enhancers for schizophrenia, modulators for OCD [4,8]) should be thought of as *tuning* the same dial in opposite directions, with the healthy brain occupying a precise, narrow window in between.

What's next

We propose several exciting directions:

- Expanding the circuit's biological realism by adding more distinct populations of excitatory and inhibitory neurons, moving beyond the current four-pool architecture toward the rich cellular diversity found in the prefrontal cortex. Incorporating these populations would make the model a much closer mirror of biology, and could reveal how the failure of specific cell types [14] contributes to the instability seen in the simulations.
- Exploring how dopamine (another key prefrontal neuromodulator) interacts with NMDA-mediated stability, particularly the D1 receptor, which is known to sharpen working memory by tuning the signal-to-noise ratio of delay-period firing [8].
- Testing whether combined NMDA + GABA reductions can reproduce the positive symptoms of schizophrenia (hallucinations, delusions) via spontaneous, unintended jumps between attractor states — a mechanism proposed by Deco & Rolls (2007)[6].
- Linking the λ_{delay} metric to measurable brain signals, fMRI delay-period activity and electrophysiology, to bridge the model with clinical data.

The bottom line

Your brain's ability to hold a thought in mind is not magic, it is physics i.e. each memory is an attractor (akin to a well in a the landscape) in the energy ladscape of brain activity. A molecular volume knob (NMDA) sets the depth of the "well" that your memory sits in. Too shallow, and the memory escapes (schizophrenia). Too deep, and it cannot be cleared (OCD). The healthy mind lives in the narrow

space between these two failure modes — and this research, for the first time, maps that space with mathematical precision.

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Code and simulation data available at: [\[github link\]](#)

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